System state check

Jaroslav Petřina

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1 System check - Assembly

```
> CALL: REG_ALL
STAT  t=1287  L=4.32  =0.86  K=5.00  C=0.47  S=0.58  emits=12
REG   RX=+0.12  RY=0.05  RZ=+0.91
ENC   BX=1.00   BY=1.30   BZ=2.10
SYM   X=1  Y=0  Z=1
GOV   =0.80  =0.50  =0.01  k=6.0  overshoot=2
INV   budget+0.03  continuity_rms=2e4  ||=0.17
LOG   [t=1279 emit =0.5 _pre=1.06, ... (last 32)]
OK
```

2 System check - Perl

```
STATUS: t=%d  L=%.2f  =%.2f  K=%.2f  C=%.2f  S=%.2f  emits:%d  |=%.2f
```

3 System check - Python

```
def get_state(self, n_log=32):
    # 1) atomic capture
    tick = self.tick_id
    L, Kcap = self.load, self.Kcap
    state = {
        "STAT": {"tick": tick, "L": L,
                 "rho_emit": (L / Kcap) if Kcap else 0.0,
                 "Kcap": Kcap, "C": self.coherence, "S": self.entropy},
        "REG":  {"RX": self.Rx, "RY": self.Ry, "RZ": self.Rz},
        "ENC":  {"BX": self.Bx, "BY": self.By, "BZ": self.Bz},
        "SYM":  {"SIGX": self.sigx, "SIGY": self.sigy, "SIGZ": self.sigz},
        "GOV":  {"lam": self.lam, "eps": self.eps,
                 "gamma": self.gamma, "k": self.k_soft,
                 "overshoot": self.overshoot_count},
        "INV":  {"budget_balance": self.work_in - (self.load + self.lambda_energy),
                 "continuity_rms": self.continuity_rms,
                 "holonomy": self.cost_holonomy},  # dict of _xy, _yz, _zx
        "LOG":  list(self.events[-n_log:])
    }
    state["CHECKSUM"] = hash(tuple(repr(v) for v in state.values()))
    return state
```

Therefore enquiry of the state of the system can be simply expressed as:

`\REG_ALL|returns STAT/REG/ENC/SYM/GOV/INV and last 32 events, as an atomic snapshot."`